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10.09.2023 MATHEMATICS

1. (1) 3

2. (4) No Solution

3. (1) 10 units

4. (2) Proportional

5. (2) $-\frac{1}{2}$

6. (1) x^3y^3

7. Let the roots of the polynomial be α and β.
ATQ,

$$\alpha + \beta = \frac{1}{2}\alpha\beta \quad \text{—— (a)}$$

We know that,

$$\alpha+\beta = \frac{-b}{a} = \frac{-(-2)}{4} = \frac{1}{2} \quad \text{—— (1)}$$

$$\alpha\beta = \frac{c}{a} = \frac{k-4}{4} \quad \text{—— (2)}$$

Putting values —(1) & —(2) in eqn —(a)

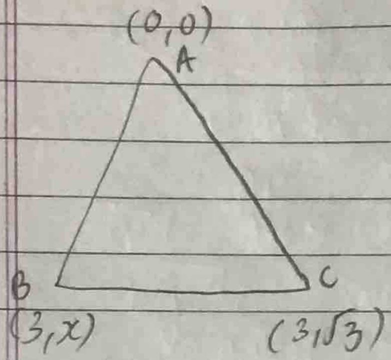
$$\cancel{\frac{1}{2}} = \cancel{\frac{1}{2}}\left(\frac{k-4}{4}\right)$$

$$1 = \frac{k-4}{4}$$

$4 = k - 4$

$8 = k$

$\boxed{k = 8}$

8. 

Given: ΔABC is equilateral

so,

$AB = AC = BC$

by Distance formula,

$AC = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$AC = \sqrt{(3-0)^2 + (\sqrt{3} - 0)^2}$

$AC = \sqrt{(3)^2 + (\sqrt{3})^2}$

$AC = \sqrt{9 + 3}$

$AC = \sqrt{12}$

$AC = 2\sqrt{3}$ units.

since $AC = AB$

so, $2\sqrt{3} = \sqrt{(3-0)^2 + (x-0)^2}$

$2\sqrt{3} = \sqrt{(3)^2 + (x)^2}$

$2\sqrt{3} = \sqrt{9 + x^2}$

by squaring both sides,

$(2\sqrt{3})^2 = (\sqrt{9 + x^2})^2$

$12 = 9 + x^2$

$3 = x^2$

$x = \pm\sqrt{3}$

9. For a unique solution,

$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

$\frac{2p}{2} \neq \frac{3}{1}$

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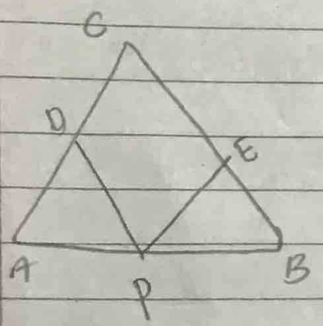
$\boxed{p \neq 3}$

Hence p can be any real value except 3.

10. Given:- $PD \parallel BC$



and $\frac{AD}{DC} = \frac{CE}{BE}$ —— (a)

To prove: $PE \parallel AC$

Proof: Since $PD \parallel BC$,
by BPT (Basic Proportionality Theorem)
$\frac{AP}{PB} = \frac{AD}{DC}$ —— (1)

Also $\frac{AD}{DC} = \frac{CE}{BE}$

so, $\frac{AP}{PB} = \frac{CE}{BE} = \frac{AD}{DC}$ [Parts equal to one another are also equal]

further, $\frac{AP}{PB} = \frac{CE}{BE}$

so, $\frac{PB}{AP} = \frac{BE}{CE}$

by converse of BPT,
$PE \parallel AC$, hence proved

11. $\tan\theta = \frac{1}{2}$ → Given

To prove: $\frac{1}{2} \sin\theta \cos\theta = \frac{1}{5}$

Proof: ~~by LHS,~~ $\tan\theta = \frac{1}{2} = \frac{P}{B}$

Let $P = 1x$, $B = 2x$

then $H = \sqrt{(1x)^2 + (2x)^2}$

$H = \sqrt{1x^2 + 4x^2}$

$H = \sqrt{5}\,x^2$

then, $\sin\theta = \frac{P}{H}$

$\sin\theta = \frac{1}{\sqrt{5}}$

and $\cos\theta = \frac{B}{H}$

$\cos\theta = \frac{2}{\sqrt{5}}$

thus, by LHS

~~$\frac{1}{2} \times \sin\theta \times \cos\theta$~~

$= \frac{1}{\cancel{2}} \times \frac{1}{\sqrt{5}} \times \frac{\cancel{2}}{\sqrt{5}}$

$= \frac{1}{5} = RHS$

since LHS = RHS, hence proved

12. $\frac{257}{5000}$

$\Rightarrow \frac{257}{2^3 \times 5^4}$

2	5000
2	2500
2	1250
5	625
5	125
5	25
5	5
	1

It was terminating decimal expansion since powers of denominarore are in form of $2^n \times 5^m$

It will terminate after 4 decimal places

13. <u>To prove:</u> $\sqrt{3}$ is an irrational number

<u>Proof:</u> let $\sqrt{3}$ be rational

so, $\sqrt{3} = \frac{a}{b}$ [where a and be are coprimes and $b \neq 0$]

by squaring both sides,

$(\sqrt{3})^2 = \left(\frac{a}{b}\right)^2$

$3 = \frac{a^2}{b^2}$

$3b^2 = a^2$

$b^2 = \frac{a^2}{3}$

3 divides a^2 then 3 must divide a —(1)

let $a = 3c$, where c is some integer

$b^2 = \frac{(3c)^2}{3}$

$b^2 = \frac{\cancel{9}^{3} c^2}{\cancel{3}}$

$b^2 = 3c^2$

$\frac{b^2}{3} = c^2$

3 divides b^2, then 3 must divide b —(2)

~~Thus Contration~~ But this contradicts the fact that a, and b are co-prime (by —(1) and —(2))

Thus Contradiction appears, this contradiction appears due to our wrong assumption

Hence $\sqrt{3}$ is an irrational number, Hence proved

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14. $\frac{1}{2\alpha-3} + \frac{1}{2\beta-3}$

$\alpha+\beta = \frac{-b}{a} = \frac{3}{2}$

$\alpha\beta = \frac{7}{2}$

$\frac{2\beta-3+2\alpha-3}{(2\alpha-3)(2\beta-3)}$

$\frac{2\beta+2\alpha-6}{4\alpha\beta-6\alpha-6\beta+9}$

$\frac{2(\alpha+\beta)-6}{4\alpha\beta-6(\alpha+\beta)+9}$

$\frac{\cancel{2}\left(\frac{3}{\cancel{2}}\right)-6}{2\cancel{4}\left(\frac{7}{\cancel{2}}\right)-\overset{3}{\cancel{6}}\left(\frac{3}{\cancel{2}}\right)+9}$

$\frac{3-6}{14-\cancel{9}+\cancel{9}}$

$\boxed{\frac{-3}{14}}$

15. For infinitely many solutions,

$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

$\frac{3}{5} = \frac{-(a+1)}{1-2a} = \frac{-2b+1}{-3b}$

$3-6a = -5(a+1)$

$3-6a = -5a-5$

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$8 = a$

$\boxed{a = 8}$ — (1)

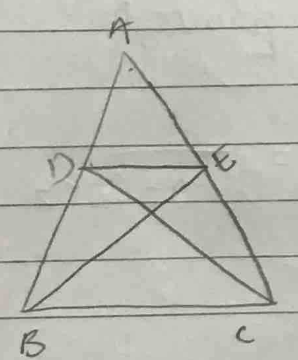
then, $\frac{-2b+1}{-3b} = \frac{3}{5}$

$-9b = -10b + 5$

$\boxed{b = 5}$ — (2)

Hence value of $a = 8$ and $b = 5$

16.



Given: $\Delta ABE \cong \Delta ACD$

$\angle A = \angle A$ — (a)

$\angle B = \angle C$ — (b)

$\angle E = \angle D$ — (c)

and $AB = AC$ — (d)

$BE = CD$ — (e)

$AE = AD$ — (f)

To prove: $\Delta ADE \sim \Delta ABC$

Proof: In ΔADE and ΔABC

$\angle A = \angle A$ (common) — (1)

and we know that,

$AB = AC$ (from (d))

and $AD = AE$ (from (f))

So, $\frac{AB}{AD} = \frac{AC}{AE}$

$\cancel{\frac{AB}{AD} - 1 = \frac{AC}{AE} - 1}$

$\cancel{\frac{AB - AD}{AD} = \frac{AC - AE}{AE}}$

~~$\frac{DB}{AD} = \frac{EC}{AE}$~~

~~so, $\frac{AD}{DB} = \frac{AE}{EC}$ — (2)~~

~~by SAS criteria~~
~~$\Delta A \sim \Delta A$~~

So, $\frac{AD}{AB} = \frac{AE}{AC}$ — (2)

by SAS criteria
$\Delta ADE \sim \Delta ABC$, hence proved

17. To prove $\frac{\tan^2\theta}{\tan^2\theta - 1} + \frac{\operatorname{cosec}^2\theta}{\sec^2\theta - \operatorname{cosec}^2\theta} = \frac{1}{\sin^2\theta - \cos^2\theta}$

Proof by LHS →

$$\frac{\tan^2\theta}{\tan^2\theta - 1} + \frac{\operatorname{cosec}^2\theta}{\sec^2\theta - \operatorname{cosec}^2\theta}$$

$$\frac{\left(\frac{\sin\theta}{\cos\theta}\right)^2}{\left(\frac{\sin\theta}{\cos\theta}\right)^2 - 1} + \frac{\frac{1}{\sin^2\theta}}{\frac{1}{\cos^2\theta} - \frac{1}{\sin^2\theta}}$$

$$\frac{\frac{\sin^2\theta}{\cos^2\theta}}{\frac{\sin^2\theta - \cos^2\theta}{\cos^2\theta}} + \frac{\frac{1}{\sin^2\theta}}{\frac{\sin^2\theta - \cos^2\theta}{\cos^2\theta \sin^2\theta}}$$

$$\frac{\sin^2\theta}{\cancel{\cos^2\theta}} \times \frac{\cancel{\cos^2\theta}}{\sin^2\theta - \cos^2\theta} + \frac{1}{\cancel{\sin^2\theta}} \times \frac{\cos^2\theta \cancel{\sin^2\theta}}{\sin^2\theta - \cos^2\theta}$$

$$\frac{\sin^2\theta}{\sin^2\theta - \cos^2\theta} + \frac{\cos^2\theta}{\sin^2\theta - \cos^2\theta}$$

by taking LCM,

$$\frac{\sin^2\theta + \cos^2\theta}{\sin^2\theta - \cos^2\theta} \quad \text{— (1)}$$

by identity,

$$\sin^2\theta + \cos^2\theta = 1$$

Putting this value in (1)

then we get,

$$\frac{1}{\sin^2\theta - \cos^2\theta} = RHS$$

Since LHS = RHS,
hence
proved.

18 $(-2,7)$ $(1,1)$ C

A 3 2 B

(x,y)

by section formula,

$$1 = \frac{mx_2 + nx_1}{m+n}$$

$$1 = \frac{3x + 2(-2)}{5}$$

$$5 = 3x - 4$$
$$9 = 3x$$
$$\boxed{x = 3} \quad — (1)$$

Also $$1 = \frac{my_2 + ny_1}{m+n}$$

$$1 = \frac{3y + 14}{5}$$

$$5 = 3y + 14$$
$$-9 = 3y$$
$$\boxed{y = -3} \quad — (2)$$

19 Let the fraction be $\frac{x}{y}$

ATQ, $$\frac{2x}{y+1} = 1 \quad — (1)$$

$$\frac{x+4}{2y} = \frac{1}{2} \quad — (2)$$

by eq^n — (1),

$$y + 1 = 2x$$
$$\boxed{2x - y = 1} \quad — (3)$$

$$2y = 2x + 8$$
$$\boxed{2x - 2y = -8} \quad — (4)$$

~~multiplying eq^n (3) by 2~~

~~4x~~

$$2x - y = 1$$
$$2x - 2y = -8$$
$$- \quad + \quad +$$

on substracting

$$\boxed{y = 9}$$

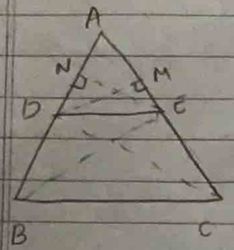
by eq^n — (3)

$$2x - 9 = 1$$
$$2x = 10$$
$$\boxed{x = 5}$$

So the fraction is $\frac{5}{9}$.

20. Basic proportionality Theorem (BPT):

If a line drawn parallel to one side of a triangle and intersects the other two sides in distinct points then the other two sides are divided in the same ration by this line.



Given: $DE \parallel BC$

To prove: $\frac{AD}{BD} = \frac{AE}{EC}$

Construction: Draw $DM \perp AC$
$EN \perp AB$
Join C to D and B to E

Proof:

Ar of $\Delta(ADE) = \frac{1}{2} \times AD \times EN$ — (1)

Ar of $\Delta(BED) = \frac{1}{2} \times EN \times DB$ — (2)

Ar of $\Delta(AED) = \frac{1}{2} \times EA \times DM$ — (3)

Ar of $\Delta(BEC) = \frac{1}{2} \times EC \times DM$ — (4)

$eq^n(1) \div eq^n(2)$

$$\frac{Ar\ of\ \Delta(ADE)}{Ar\ of\ (\Delta BED)} = \frac{\cancel{\frac{1}{2}} \times AD \times \cancel{EN}}{\cancel{\frac{1}{2}} \times \cancel{EN} \times DB}$$

$$= \frac{AD}{DB} \quad \text{— (5)}$$

$eq^n(3) \div eq^n(4)$

$$\frac{Ar\ \Delta AED}{Ar\ \Delta BEC} = \frac{\cancel{\frac{1}{2}} \times EA \times \cancel{DM}}{\cancel{\frac{1}{2}} \times EC \times \cancel{DM}}$$

$$= \frac{EA}{EC} \quad \text{— (6)}$$

Since ΔBDE and ΔEC are on same base (DE) and b/w same parallel lines $(DE \parallel BC)$

so, $Ar\ \Delta(BDE) = Ar\ \Delta(DEC)$

by — (5) and — (6)

$$\frac{Ar\ (\Delta ADE)}{Ar\ (\Delta BDE)} = \frac{Ar\ (\Delta ADE)}{Ar\ (\Delta DEC)}$$

$\frac{AD}{BD} = \frac{AE}{EC}$, hence proved.

21. $$\frac{\sin 30^\circ + 2\cos^2 45^\circ + \tan^2 60^\circ}{\frac{1}{2}\cot 45^\circ + \cos^2 30^\circ + \tan^2 45^\circ}$$

$$\frac{\frac{1}{2} + 2\left(\frac{1}{\sqrt{2}}\right)^2 + (\sqrt{3})^2}{\frac{1}{2} \times 1 + \left(\frac{\sqrt{3}}{2}\right)^2 + (1)^2}$$

$$\frac{\frac{1}{2} + \cancel{2}\left(\frac{1}{\cancel{2}}\right) + 3}{\frac{1}{2} + \frac{3}{4} + 1}$$

$$\frac{\frac{1+2+6}{2}}{\frac{2+3+4}{4}} = \frac{\frac{9}{2}}{\frac{9}{4}}$$

$$= \frac{\cancel{9}}{\cancel{2}} \times \frac{\cancel{4}^{\,2}}{\cancel{9}}$$

$$\boxed{= 2}$$

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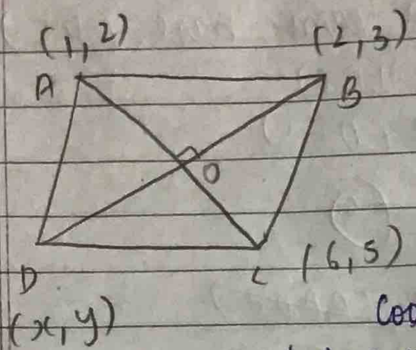
22.



We know that,

~~AD = BC~~

diagonals of parallelogram bisect each other,

$\left(\frac{x_1+x_2}{2}\right), \left(\frac{y_1+y_2}{2}\right)$

Coordinates of point O $= \frac{1+6}{2}, \frac{2+5}{2}$

$= \frac{7}{2}, \frac{7}{2}$

and also, $\left(\frac{7}{2}, \frac{7}{2}\right) = \frac{2+x}{2}, \frac{y+3}{2}$

$\frac{7}{\cancel{2}} = \frac{2+x}{\cancel{2}}$

$\boxed{5 = x}$

$\frac{7}{\cancel{2}} = \frac{y+3}{\cancel{2}}$

$\boxed{y = 4}$

23. A will complete 1 round in $= \frac{198\cancel{0}}{33\cancel{0}}$

$= 6$ min

B " " " " " $= \frac{1980}{198}$

$= 10$ min

C " " " " " $= \frac{198\cancel{0}}{22\cancel{0}}$

$= 9$ min

req no = LCM (6, 10, 9)

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$6 = 2 \times 3$
$10 = 2 \times 5$
$9 = 3^2$

$LCM(6, 10, 9) = 2 \times 3^2 \times 5$
$= 10 \times 3^2$
$= 10 \times \cancel{3} 9$
$= 90$

After 90 min they will be together at the starting point.

24. son's age – x years
father's age – y years

ATQ, $2x + y = 70$ —(1)
$x + 2y = 95$ —(2)

eq (1) $\times 2$

$4x + \cancel{2y} = 140$
$x + \cancel{2y} = 95$
$-\quad -\quad -$

on substracting

$3x = 45$
$\boxed{x = 15}$ years

$15 + 2y = 95$
$2y = 80$
$\boxed{y = 40}$ years

son's age = 15 years
father's age = 40 years.

25. Any point on x axis $= (x, 0)$

$(x, 0)$

$(1,-7) \quad k \quad 1 \quad (6,4)$

Let the ratio be $\boxed{k:1}$

$$0 = \frac{m y_2 + n y_1}{m+n}$$

$$0 = \frac{4k + (-7)}{k+1}$$

$$k + 1 = 4k - 7$$

$$k + 8 = 4k$$

$$8 = 3k$$

$$k = \frac{8}{3}$$

$$k : 1 = \frac{8}{3} : 1$$

Ratio $\Rightarrow \boxed{8:3}$

26. Given: $\operatorname{cosec}\theta - \sin\theta = m$

$\sec\theta - \cos\theta = n$

$\rightarrow m^2 n$

$$(\operatorname{cosec}\theta - \sin\theta)^2 (\sec\theta - \cos\theta)$$

$$\left(\frac{1}{\sin\theta} - \sin\theta\right)^2 \left(\frac{1}{\cos\theta} - \cos\theta\right)$$

$$\left(\frac{1-\sin^2\theta}{\sin\theta}\right)^2 \left(\frac{1-\cos^2\theta}{\cos\theta}\right)$$

$$\cancel{\left(\frac{1-\sin^4\theta}{\sin^2\theta}\right)\left(\frac{1-\cos^2\theta}{\cos\theta}\right)}$$

$$\cancel{\sin^2\theta \cos^2\theta}$$

$$\left(\frac{\cos^2\theta}{\sin\theta}\right)^2 \left(\frac{\sin^2\theta}{\cos\theta}\right) \quad \because \sin^2\theta + \cos^2\theta = 1$$

$$\frac{\cos^4\theta \cancel{\sin^2\theta}}{\cancel{\sin^2\theta} \cos\theta}$$

$$\cos^3\theta$$

then $(m^2 n)^{\frac{2}{3}} = (\cos^3\theta)^{\frac{2}{3}}$

$$= \cos^2\theta$$

$$mn^2 = (\operatorname{cosec}\theta - \sin\theta)(\sec\theta - \cos\theta)^2$$

$$= \left(\frac{1}{\sin\theta} - \sin\theta\right)\left(\frac{1}{\cos\theta} - \cos\theta\right)^2$$

$$= \left(\frac{1-\sin^2\theta}{\sin\theta}\right)\left(\frac{1-\cos^2\theta}{\cos\theta}\right)^2$$

$$= \left(\frac{\cos^2\theta}{\sin\theta}\right)\left(\frac{\sin^2\theta}{\cos\theta}\right)^2$$

$$= \frac{\cancel{\cos^2\theta} \sin^4\theta}{\sin\theta \cancel{\cos^2\theta}}$$

$$= \sin^3\theta$$

then $(mn^2)^{2/3} = (\sin^3\theta)^{\frac{2}{3}}$

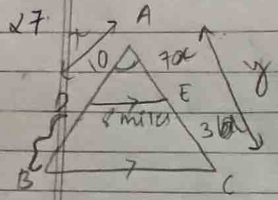
$$= \sin^2\theta$$

them to prove, $(m^2 n)^{2/3} + (mn^2)^{2/3} = 1$

$\cos^2\theta + \sin^2\theta$

1

LHS = RHS, hence proved.

27 

A → Mountain top

D → hut 1

E → hut 2

B → Hotel A

C → Hotel B

then we know, $\frac{10}{DB} = \frac{7}{3}$ by BPT,

$\cancel{30 = 7DB}$

$\cancel{\frac{30}{7} = DB}$

$$\frac{AD}{DB} = \frac{AE}{EC}$$

$$\frac{DB}{AD}+1 = \frac{EC}{AE}+1$$

$$\frac{AB}{AD} = \frac{AC}{AE} \quad \text{— (1)}$$

and $\angle A = \angle A$

$\triangle ABC \sim \triangle ADE$

then,

i) $\frac{AD}{AB} = \frac{AE}{EC} = \frac{DE}{BC} = \frac{3}{10}$

$$\frac{AD + AE + DE}{AB + EC + BC} = \frac{9}{30} = \frac{3}{10}$$

$\boxed{3:10}$ — Ratio of Perimeter.

ii) since by BPT

$$\frac{10}{DB} = \frac{7}{3}$$

$30 = 7DB$

$DB = \frac{30}{7}$

$DB = 4.2$ miles.

iii) $\frac{8}{BC} = \frac{3}{10}$

$80 = 3BC$

$BC = \frac{80}{3}$

$BC = 26.6$ miles.

iv) $x = 5 + y$

$10 = 5 + 10a$

$5 = 10a$

$a = \frac{1}{2}$

$AE = 7\left(\frac{1}{2}\right)$

$AE = \frac{7}{2}$

$AE = 3.5$ miles.

28 P (2, 5)

Q (x, y)

R (8, 3)

ii) $x = \frac{mx_2 + nx_1}{m+n}$

$x = \frac{8+4}{3}$

$3x = 12$

$\boxed{x = 4}$

$y = \frac{my_2 + ny_1}{m+n}$

$y = \frac{3+10}{3}$

$3y = 13$

$y = \frac{13}{3}$

$\boxed{y = 4.3}$

Coordinates of $Q = (4, 4.3)$

$= \left(4, \frac{13}{3}\right)$

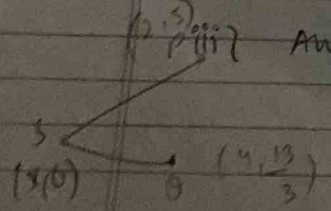
ii) $PR = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

$= \sqrt{(8-2)^2 + (3-5)^2}$

$= \sqrt{(6)^2 + (-2)^2}$

$= \sqrt{36+4}$

$= \sqrt{40}$ units



iii) Any point on x axis → $(x, 0)$

$\sqrt{(2-x)^2 + (5-0)^2} = \sqrt{(4-x)^2 + \left(\frac{13}{3} - 0\right)^2}$

After Squaring both sides,

$(2-x)^2 + 25 = (4-x)^2 + \frac{169}{9}$

$4 + \cancel{x^2} - 4x + 25 = 16 + \cancel{x^2} - 8x + \frac{169}{9}$

$4x + 25 = 12 + \frac{169}{9}$

$4x = 13 + \frac{169}{9}$

$4x = \frac{117 + 169}{9}$

$36x = 286$

$x = \frac{286}{36} = \frac{143}{18}$

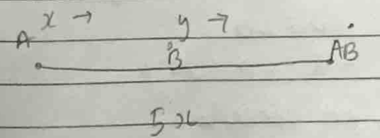
Coordinates are $\left(\frac{143}{18}, 0\right)$

iv) $(x, y) = \left(\frac{2+8}{2}\right), \left(\frac{5+3}{2}\right)$

$= \left(\frac{10}{2}\right), \left(\frac{8}{2}\right)$

$= (5, 4)$

29.



Let the speed of Car A = x Km/hr

Let the speed of Car B = y Km/hr

distance covered by A in 5 hours = 5x km

distance covered by B in 5 hours = 5y km

ATQ,

$5x - 5y = 100$

$\boxed{x - y = 20}$ — ①

A 100 km B

Distance covered by car A in 1 hr = x k
Distance covered by car B in 1 hr = y k

By Question,

$$\boxed{x+y=100} \quad \text{--- (2)}$$

by elimination method,

$$x-\cancel{y}=20$$

on adding

$$x+\cancel{y}=100$$

$$2x=120$$

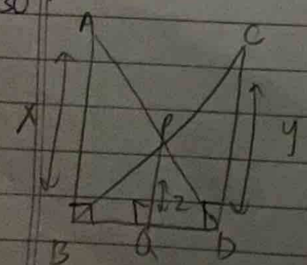
$$x=60 \text{ km/hr}$$

by eqn (2) $60+y=100$

$$y=40 \text{ km/hr}$$

Thus, speed of car A = 60 km/hr
and speed of car B = 40 km/hr.

30.



In ΔBPQ and ΔBCD,

$\angle B = \angle B$

$\angle Q = \angle D \ (90°)$

by AA criteria

$\Delta BQP \sim \Delta BDC$ --- (1)

In ΔDPQ and ΔDAB

$\angle D = \angle D$

$\angle Q = \angle B \ (90°$

by AA criteria

$\Delta DQP \sim \Delta DBA$ --- (2)

by (1), $\frac{PQ}{CD} = \frac{z}{y} = \cancel{BQ}\ \frac{BQ}{DB}$

by (2) $\frac{PQ}{AB} = \frac{z}{x} = \frac{DQ}{DB}$

Adding (1) and (2)

$$\frac{z}{y}+\frac{z}{x} = \frac{BQ+DQ}{DB}$$

$$\frac{z}{y}+\frac{z}{x} = \frac{DB}{DB}$$

$$z\left(\frac{1}{y}+\frac{1}{x}\right) = 1$$

$$\boxed{\frac{1}{y}+\frac{1}{x} = \frac{1}{z}}$$, hence proved.